

Ramesh Vawadra

Firmware Developer

PROFESSIONAL EXPERIENCE

- A dynamic professional with 4+ years of experience in the areas of Firmware development and testing. Adept at designing and building applications with usability and high performance in mind.

CAREER HIGHLIGHTS:

Primary Skill	C, Embedded C, Python, GCC, STM32 IDE, IAR, Segger, Linux, HLOS, nonHLOS, UEFI, gdb, debugger, Logic Analyzers, Tests Validation
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- Experience in working STM32 , nrf MCUs and Qualcomm linux systems, ARM M architectures including industrial electronic products
- Extensive Full Cycle experience in embedded system development and implementation including System architecture, Firmware development, testing, debugging and documentation
- Experience in Embedded RTOS (FreeRTOS), working knowledge of communication protocols SPI, UART, I2C, QSPI, NVME, UFS, EMMC, PCIE
- Experience in unit test cases , test strategies and scenarios
- Self-Starter, Self-Motivated and Quick Learner. Team Player with Good Communication and Problem-Solving skills.

PROJECT EXPERIENCE

Dimension NXG / Firmware Developer, March 2023 onwards

AR Head Mounted Display

- Description and Features: FreeRTOS-based Head Mounted Display is a passthrough display device with inbuilt 9DoF IMU catering to Defense applications
- Developed firmware for USB, IMU, and peripheral integration on STM32F4 series MCU
- Validated system functionality and performance, including troubleshooting and debugging
- Integrated multiple IMUs to achieve precise accuracy requirements, validated algos to achieve the requirements

BLE based Motion Sensing Device

- Description and Features: Motion device senses the movement, with in specific tolerance catering to Defense applications
- Developed calibration features and utilized Python scripting for automation
- Firmware software implementation for BLE motion sensing applications
- Optimized data structures to efficiently manage inputs from multiple IMUs
- Collaborated on defining device specifications and conducted thorough validation

POC: Pressure sensing Gloves

- Description and Features: Palmar Pressure feedback gloves for VR Simulation
- Developed POC for sensing pressure using resistive , capacitive sensors pads
- Implemented ADC for multiple sensor pads
- Debug and developed for acquisition of ADC readings added in library (outdated lib)

Acevin Solutions / Firmware Engineer, Dec 2020 - March 2023

OTS-SSM

- Description and Features: The Spotlight Mote is an IOT device that consists of a WSCB (Wireless Sensor Communication Board) and ASB (Analog Sensor Board).
- ASB does all the analog sensor signal conditioning and digitization. Based on the sensor types WSCB configures the ASB, collects/ process the data, and transmits the information wirelessly to the gateway.
- The WSCB has two processors (Nordic and DUST or SmartMesh IP).
- The Spotlight mote is powered by a non-rechargeable battery, typically having a minimum of one life when operated in a 25 C environment. Each Mote in the network are synchronized with the network Manager (Gateway) and will periodically receive commands from a manager to wake up and collect sensor readings.
- Signal Processing is performed on the data to provide Static values and/or waveform information.
- QSPI interfaced external PS RAM for storing large chunks of raw and signal processed ADC samples
- Power optimization of device in awake and deep sleep states as a result achieving the battery life of more than 1 year

EMsave ILT

- Description and Features: EMSave ILT is a Integrated Lamp Test controller board, It has functions to test the peripherals of the lamps and assigning the characteristics info upto 120 lamps at a time.
- Controller board consists of STM32f303 which receives data from RaspberryPi via UART.
- Developed driver for Uart based communication between RPI and STM32f303 micro controller
- Developed driver for I2C based IO expanders for driving 120 relays with Single and multiple at instance

Here Technologies / Data Engineer, March 2019 - Dec 2020

AirMouse

- Description and Features: AirMouse was a inhouse prototyped mouse version for interacting with tools enabling more robust workflow between users
- Developed a prototype version of Air ring mouse ,based on the accelerometers adxl345 and STM32 micro controller
- Designed and 3d enclosure for the ring prototype version
- Validation of frame functionality of the device firmware

EarthScape workflow

- Handled the project right from requirement gathering, stackholder interaction, report generation and analysis and application performance monitoring.
- Defect log using Project Management tool like Confluence.
- Logging the defect with the proper logs, screenshots and tracking of the defect.

EDUCATION

Bachelor of Engineering in Electronics Telecomm., Mumbai University,

Pillai HOC College of Engineering

June 2017

HSC Board, Maharashtra State Board, Sainath Junior College

March 2013

SSC Board, Maharashtra State Board, Tilak High School

March 2011

PERSONAL DETAILS

Full Name : Ramesh Vawadra

Date of Birth : 23rd Sept 1994

Hobbies : Photography , Painting

Address : Navi Mumbai, Maharashtra - 400703

DECLARATION

I solemnly declare the information mentioned herein is true and correct to the best of my beliefs.

Regards,

Ramesh Vawadra